

OLS Regression

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Problem

You are provided with a dataset containing the information about a small business's monthly advertising expenditure and corresponding sales. The dataset includes the following variables:

- Advertising Budget (X):** the amount spent on advertising, in **thousands of dollars**
- Sales (Y):** the number of units sold, in **thousands**

Month	Advertising Budget (X)	Sales (Y)
1	3.0	6.0
2	5.0	9.0
3	2.0	4.0
4	7.0	10.0
5	8.0	12.0
6	1.0	3.0
7	4.0	7.0
8	6.0	8.0
9	9.0	13.0
10	10.0	15.0

```
In [22]: import numpy as np

In [23]: # Problem
x = np.array([3., 5., 2., 7., 8., 1., 4., 6., 9., 10.])
y = np.array([6., 9., 4., 10., 12., 3., 7., 8., 13., 15.])

In [24]: # Check the amount of dataset
n = len(x)
print(f"n = {n}")

n = 10
```

Task 1 Implement OLS Regression

Method 1: Using the OLS method

Fit $\hat{y} = \alpha + \beta x$

$$\hat{\beta} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}, \quad \hat{\alpha} = \bar{y} - \hat{\beta} \bar{x}$$

```
In [25]: # Compute mean value
x_bar = np.mean(x)
y_bar = np.mean(y)

# Compute different value from mean
dx = x - x_bar
dy = y - y_bar

# Sum( (x_i - x_bar)(y_i - y_bar) )
s_xy = np.sum(dx * dy)

# Sum(x_i - x_bar)^2
s_xx = np.sum(dx ** 2)

# Final formular
beta = s_xy / s_xx
alpha = y_bar - beta * x_bar

In [26]: print(f"x_bar = {x_bar}      y_bar = {y_bar}")
print(f"S_xy = {s_xy}      S_xx = {s_xx}")
print(f"intercept = {alpha:.4f}")
print(f"slope = {beta:.4f}")

x_bar = 5.5      y_bar = 8.7
S_xy = 104.5      S_xx = 82.5
intercept = 1.7333
slope = 1.2667
```

Method 2: using matrix normal equation (optional)

```
In [27]: # matrix A -> [[1, x_1], [1, x_2], [1, x_3], ... ]
# [1, ...] is alpha term, while [..., x_1] is beta term
# n = 10
A = np.column_stack((np.ones(n), x))

# Pseudo-inverse -> AB = y -> B = A* y
B = np.linalg.pinv(A) @ y

In [28]: print("A.shape =", A.shape)
print(f"B = {B}")
print(f"intercept = {B[0]:.4f}")
print(f"slope = {B[1]:.4f}")

A.shape = (10, 2)
B = [1.73333333 1.26666667]
intercept = 1.7333
slope = 1.2667
```

Summary (Final Answer)

$\hat{\alpha} = 1.7333$ and $\hat{\beta} = 1.2667$.

$$\hat{y} = 1.7333 + 1.2667x$$

Task 2 — Interpret the Results

```
In [29]: x_new = 12.0
y_new = alpha + beta * x_new

In [30]: print(f"budget = ${x_new * 1000:,.0f}")
print(f"predicted sales = {y_new:.4f} thousand units = {y_new * 1000:,.0f} units")

budget = $12,000
predicted sales = 16.9333 thousand units = 16,933 units
```

Explain the meaning of the slope and intercept in the context of this problem.

Slope: ($\hat{\beta} = 1.2667$): For every additional \$1,000 spent on advertising (X), the model predicts sales to increase by about 1.2667 thousand units (Y).

Intercept: ($\hat{\alpha} = 1.7333$): When advertising spending between 0 (X), the model predicts about 1.7333 thousand units sold (Y). It remains constant and does not change along with advertising (X).

Limitation: this is the observed range of X (1 to 10), The Outside the range of 1 to 10 isn't promise.

Predict the sales if the advertising budget is \$12,000.

At $x = 12$ (thousand units), the predicted sales are about 16.9333 thousand units, or about 16,933 units. This is an extrapolation because 12 is outside the observed range.

Task 3 — Model Evaluation

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}}$$

```
In [31]: # Predict value
y_hat = alpha + beta * x

# Our model predict score
ss_res = np.sum((y - y_hat) ** 2)

# Average predict score
ss_tot = np.sum((y - y_bar) ** 2)

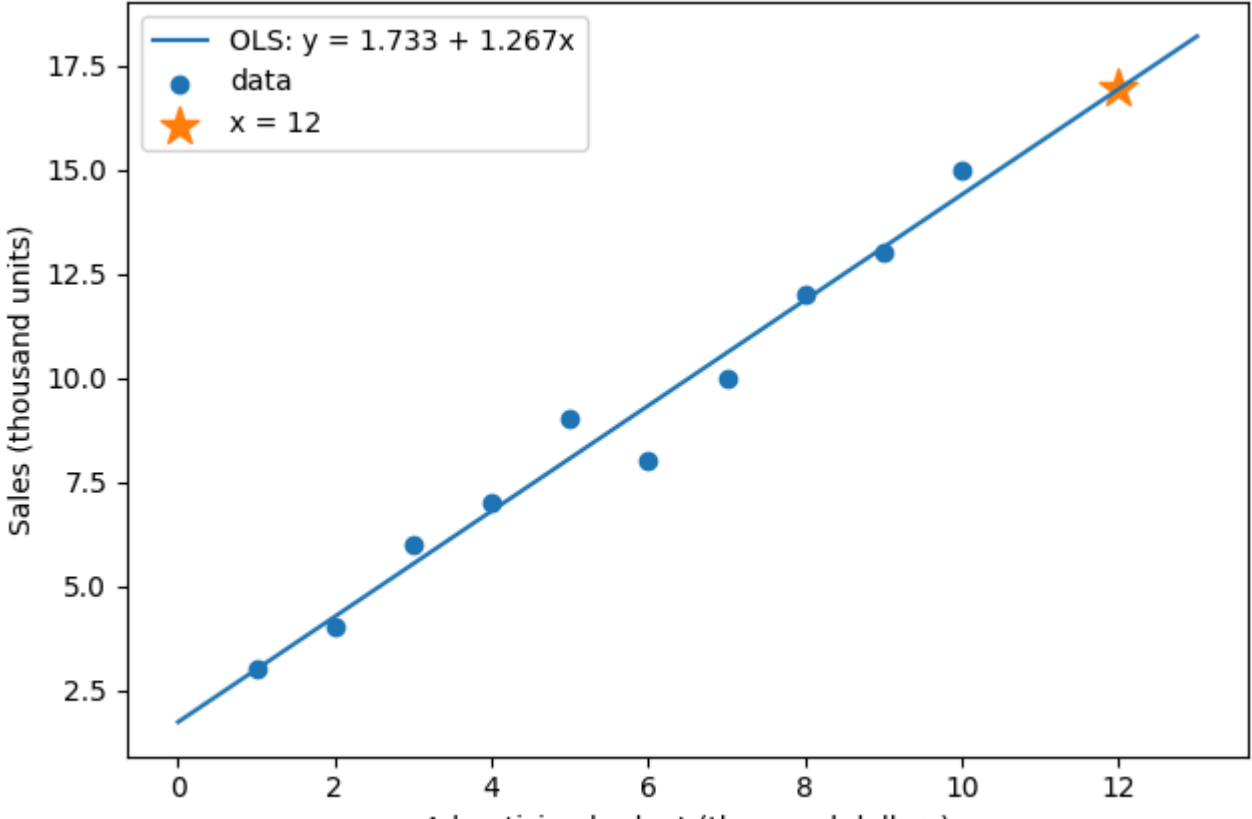
# r square compute
r2 = 1 - ss_res / ss_tot

In [32]: print(f"SS_res = {ss_res:.4f}")
print(f"SS_tot = {ss_tot:.4f}")
print(f"R^2 = {r2:.4f}")

SS_res = 3.7333
SS_tot = 136.1000
R^2 = 0.9726

In [33]: import matplotlib.pyplot as plt

grid = np.linspace(x.min() - 1, x_new + 1, 100)
plt.figure(figsize=(6.5, 4.5))
plt.plot(grid, alpha + beta * grid, label=f"OLS: y = {alpha:.3f} + {beta:.3f}x")
plt.scatter(x, y, label="data")
plt.scatter([x_new], [y_new], marker="*", s=200, label="x = 12")
plt.xlabel("Advertising budget (thousand dollars)")
plt.ylabel("Sales (thousand units)")
plt.legend()
plt.tight_layout()
plt.show()
```



Provide a brief interpretation of the R-squared value obtained.

$R^2 = 0.9726$ means that roughly 97.26% of the variation in sales can be explained by the advertising budget, which indicates a strong fit for the model. The remaining ~3% comes from other unmeasured factors or random noise.

Keep in mind that a high R^2 shows a strong correlation, but it doesn't prove that advertising directly causes the increase in sales.